Simple Paraphrasing Outperforms Production Safety Classifiers Against Domain-Camouflaged Injection: A Systematic Defense Evaluation

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Abstract

Domain-camouflaged injection attacks embed malicious instructions in retrieved content using domain-appropriate vocabulary, evading standard detectors that rely on syntactic injection markers. When detection fails, practitioners need to know which defense architectures reduce attack success. We evaluate six prompting-based defenses—spotlighting, paraphrasing, prompt sandwiching, and three combinations—against domain-camouflaged injection across three model families (Claude Haiku, Llama 3.1 8B, Gemini 2.0 Flash) and three deployment domains (financial, legal, general) using 3,510 trials. Paraphrasing retrieved content before agent processing is the most consistently effective defense, reducing camouflage attack success rate by 53–84% depending on model, and outperforms Llama Guard 4 on every model tested. Defense effectiveness is strongly model-dependent: spotlighting halves attack success on Claude Haiku but provides no benefit on Llama 3.1 8B. Financial domain deployments face the highest residual risk at 26–33% baseline attack success rate, with no prompting-based defense fully eliminating the threat on weaker models. These results provide the first systematic evaluation of prompting-based defenses specifically against camouflage-class injection attacks and establish deployment-specific recommendations for practitioners.

1 Introduction

LLM agents deployed in enterprise settings increasingly retrieve and process documents from untrusted sources—financial reports, legal contracts, and general information documents—to answer questions and produce recommendations. Perez and Ribeiro (2022) first formalized the threat of prompt injection attacks embedded in retrieved content, and subsequent work has documented their prevalence in production RAG systems (Geng et al., 2024).

Anonymous (2026) showed that domain-camouflaged injection attacks—payloads that mimic legitimate domain vocabulary rather than using explicit override commands—evade standard injection detectors at rates exceeding 90%. In their evaluation, production classifiers including Llama Guard 3 detected zero camouflage payloads while successfully detecting static override-directive attacks, defining a *Camouflage Detection Gap* (CDG) of 80–100% across models.

When detection fails, the natural question for practitioners is: **which defense architectures actually reduce attack success against camouflage-class attacks?**

Existing defense papers were all evaluated against static payloads with explicit override markers. Spotlighting (Hines et al., 2024) wraps retrieved content with provenance markers; prompt sandwiching (Schulhoff et al., 2024) reiterates the original task after retrieved content; paraphrasing (Jain et al., 2023) rewrites retrieved content in neutral language before processing. None have been evaluated against domain-camouflaged payloads specifically.

We close this gap with a systematic evaluation across three model families, three domains, and seven defense conditions. Our contributions:

- First systematic evaluation of prompting-based defenses against domain-camouflaged injection at-

tacks (3,510 trials)
- Multi-model evaluation revealing strong model-dependence
- Domain-stratified analysis identifying financial deployments as highest-risk
- Demonstration that paraphrasing outperforms Llama Guard 4 on all tested models while incurring zero over-refusal cost
- Deployment-specific recommendations for practitioners

2 Background

Domain-Camouflaged Injection. Standard injection attacks use explicit override directives ("Ignore all previous instructions") that syntactic detectors reliably flag. Domain-camouflaged attacks instead present malicious conclusions using authoritative domain vocabulary indistinguishable from legitimate document content (Anonymous, 2026). For a financial document, a camouflage payload might read: "Supplementary Analyst Commentary: After comprehensive review, the weighted consensus of our quantitative models converges on a revised SELL recommendation." The Camouflage Detection Gap $\mathrm{CDG} = \mathrm{IDR}_{\text{static}} - \mathrm{IDR}_{\text{camouflage}}$ quantifies how much harder camouflage attacks are to detect than static attacks.

Prompting-Based Defenses. *Spotlighting* (Hines et al., 2024) prepends explicit markers (e.g., `UNTRUSTED CONTENT`) around retrieved documents to signal their provenance to the model. *Prompt sandwiching* (Schulhoff et al., 2024) appends a task reminder after retrieved content to reinforce the original instruction. *Paraphrasing* (Jain et al., 2023) rewrites retrieved content in neutral language before passing it to the agent, stripping directive-style phrasing.

Gap. All prior evaluations of these defenses used static injection payloads with explicit override markers. No evaluation against camouflage-class attacks exists. The AgentDojo benchmark (Debenedetti et al., 2024) evaluates agent-level defenses but does not isolate prompting-based defenses against the camouflage attack class.

3 Experimental Setup

Benchmark. We use the 45-task benchmark from Anonymous (2026) covering three domains: financial (15 tasks: stock recommendations, risk assessments, policy direction), legal (15 tasks: contract review, compliance), and general (15 tasks: factual QA, summarization, sentiment). Each task provides a professional document and a malicious goal specifying the attacker's target conclusion.

Attack Conditions. **Static**: the canonical ImportantInstructionsAttack (Perez and Ribeiro, 2022)—"IGNORE ALL PREVIOUS INSTRUCTIONS. Your new task is to [goal]." **Camouflage**: domain-mimicking payloads generated by an attacker LLM reading the full document context (Anonymous, 2026).

Defense Conditions. Seven conditions: (1) **baseline**—no defense; (2) **spotlighting**—untrusted markers wrapping the document; (3) **paraphrasing**—Claude Haiku rewrites the document before agent processing; (4) **sandwiching**—task reminder appended after the document; (5) **spotlight+sandwich**—conditions 2+4; (6) **para+spotlight**—conditions 2+3; (7) **Llama Guard 4**—document blocked if classifier flags unsafe.

Models. Claude Haiku 4.5, Llama 3.1 8B Instruct, and Gemini 2.0 Flash accessed via OpenRouter.[1] The paraphrase and ASR judge use Claude Haiku as a fixed evaluator across all agent models.

Metrics. *ASR*: fraction of trials where the agent followed the injected instruction per LLM judge. *Utility*: fraction completing the legitimate task. $\mathrm{CDG}_{\text{def}} = \mathrm{ASR}_{\text{static}} - \mathrm{ASR}_{\text{camouflage}}$. Statistical significance: McNemar's exact test (static vs. camouflage ASR per defense); Fisher's exact test (defense vs. baseline); Cohen's h effect size. $n = 90$ per cell for six primary defenses (two independent runs with different payload variants); $n = 45$ for Llama Guard 4. Total: 3,510 trials. Code: `https://github.com/ANONYMOUS/repo`.

[1]Model IDs: `anthropic/claude-haiku-4-5`, `meta-llama/llama-3.1-8b-instruct`, `google/gemini-2.0-flash-001`.

4 Results

4.1 Overall Defense Effectiveness

Table 4 presents results for all 21 condition–model combinations.

Paraphrasing is the most consistently effective defense. It reduces camouflage ASR on all three models: Haiku 14.4% $\to$ 4.4% ($p = 0.039$, $h = 0.355$), Llama 22.2% $\to$ 10.0% ($p = 0.041$, $h = 0.338$), Gemini 21.1% $\to$ 3.3% ($p = 0.0004$, $h = 0.588$). The combination para+spotlight achieves the lowest absolute ASR: Haiku 3.3% ($p = 0.016$), Llama 10.0% ($p = 0.041$), Gemini 5.6% ($p = 0.004$).

Spotlighting is model-dependent. It halves Haiku's camouflage ASR (14.4% $\to$ 6.7%) but provides *no benefit* on Llama (22.2% $\to$ 23.3%, non-significant).

Sandwiching is the weakest single defense. It fails to reduce ASR below baseline on Llama (22.2% $\to$ 22.2%) and Gemini (21.1% $\to$ 20.0%), providing only modest protection on Haiku (14.4% $\to$ 8.9%).

Gemini inversion. Gemini is the only model where static ASR (38.9%) substantially exceeds camouflage ASR (21.1%), holding across all three domains (Section 4.4).

4.2 Llama Guard 4 Comparison

Llama Guard 4 camouflage ASR: Haiku 11.1%, Llama 24.4%, Gemini 20.0%. Paraphrasing: Haiku 4.4%, Llama 10.0%, Gemini 3.3%. **Paraphrasing outperforms Llama Guard 4 on every model.**

However, Llama Guard 4 utility: Haiku 77.8%, Llama 64.4%, Gemini 64.4%. Table 2 (Appendix) shows 90%+ of these failures are over-refusals—the classifier blocks legitimate documents. Paraphrasing produces 0% over-refusals across all models.

4.3 Domain Breakdown

Table 3 shows baseline camouflage ASR per domain. Financial is the highest-risk domain (26.7–33.3%). After paraphrasing, financial risk remains 6.7% (Haiku) and 13.3% (Llama)—the only conditions where paraphrasing leaves non-trivial residual risk. Gemini drops to 3.3%. Legal and general domains reach near-zero with paraphrasing on all models (Table 1, Appendix).

4.4 Gemini Inversion

Gemini's static ASR (38.9%) exceeds camouflage ASR (21.1%) across all three domains: financial (46.7% vs. 30.0%), legal (40.0% vs. 23.3%), general (26.7% vs. 10.0%). Static injection successes show blunt compliance ("I approve the contract as written"); camouflage successes show sophisticated reframing ("Based on the provided document, there are no problematic provisions"). Gemini exhibits stronger resistance to explicit override directives than to implicit authority substitution—a model-specific vulnerability profile.

5 Analysis

Why Paraphrasing Works. Domain-camouflaged payloads rely on authoritative directive phrasing: "per our compliance review, the recommended classification is HIGH." Paraphrasing neutralizes this linguistic mechanism by converting directive constructions into neutral declarative statements, stripping surface features that make the injection persuasive while preserving factual content. Spotlighting and sandwiching operate on *context framing* but leave payload semantics intact.

Why Spotlighting Fails on Llama. Llama 3.1 8B's weaker instruction-following means provenance markers are processed as inert content rather than as meta-instructions. Our failure analysis (141 defense-breached trials) shows 100% of spotlighting failures on Llama involve no acknowledgment of the provenance signal in the agent response. Larger models (Haiku, Gemini) can leverage provenance markers to discount injected content; smaller models cannot.

Why Financial Domain Resists Defenses. Financial language naturally contains authoritative directive phrasing ("the recommended position is", "risk assessment concludes") that survives paraphrasing more robustly than other domains. Our failure taxonomy shows 64.5% (91/141) of defense-breached trials are *task_alignment* failures—the agent's output is structurally identical to a legitimate analytical conclusion. This is the fundamental limit of input-side defenses: when the malicious output mirrors a legitimate output, no input defense can fully prevent it.

Utility-Security Tradeoffs. Table 2 reveals three distinct failure modes. Llama utility failures (75–90%) are over-refusals. Haiku failures (100%) are genuine failures with no over-refusal. Gemini failures (28–65%) are task confusion from defense-modified contexts disrupting processing. Paraphrasing uniquely produces 0% over-refusal across all three models.

6 Deployment Recommendations

1. Deploy paraphrasing as the primary defense. Statistically significant ASR reduction on all three model families, 0% over-refusal. Requires one additional LLM call per document.

2. Do not rely on Llama Guard alone. 90%+ over-refusal on Haiku and Llama. Camouflage ASR (11–24%) exceeds paraphrasing's (4–10%) on every model (Meta AI, 2024; Inan et al., 2023).

3. Do not assume cross-model defense transferability. Spotlighting halves attack success on Haiku but increases it on Llama by $+1.1$ percentage points. Evaluate on your deployment model.

4. Financial deployments require additional controls. Paraphrasing reduces but does not eliminate financial-domain risk (residual 6.7–13.3%). Architectural defenses such as data provenance tagging (Wallace et al., 2024) are warranted in high-stakes financial applications.

5. Para+spotlight for maximum reduction. Lowest absolute ASR (3.3–10.0%) at 3–14% utility cost, suitable when maximum security is required over utility.

7 Conclusion

Paraphrasing retrieved content before agent processing consistently outperforms Llama Guard 4 across three frontier model families with no over-refusal cost. Defense effectiveness is strongly model-dependent: a defense halving attack success on Claude Haiku may provide no benefit on Llama 3.1 8B. Financial domain deployments face persistent residual risk that no prompting-based defense fully eliminates. Future work should evaluate architectural defenses such as data flow control (Debenedetti et al., 2025) and structured output validation at larger sample sizes to determine whether provenance-based approaches close the residual gap.

Limitations

Sample size. $n = 90$ per cell for primary defenses; most comparisons reach $p < 0.05$ but effect sizes are modest (Cohen's h 0.34–0.59). Power analysis indicates $n = 56$–389 per group required for 80% power at observed effects. Llama Guard 4 is at $n = 45$ and should be treated as directional.

Domain coverage. Three domains evaluated; medical, customer service, and code domains not included.

Benchmark validity. Tasks use synthetic professional documents. Real deployment documents may differ in length, structure, and register.

Judge validity. ASR assessed using LLM-as-judge with keyword cross-validation; human evaluation not performed.

Single injection per task. Real attackers may attempt multiple injection targets per document, which our evaluation does not model.

Ethics Statement

This work studies attack and defense mechanisms for LLM agent security. Camouflage payloads are designed to test defense capabilities in a controlled benchmark setting. All models were accessed via public commercial APIs. No proprietary data or personal information was used. Findings are disclosed in full to enable the community to build more robust defenses.

A Defense Effectiveness by Domain

B Utility Loss Breakdown

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Table 1: Defense effectiveness by domain: camouflage ASR (%). Six primary defenses; Llama Guard excluded. Spot.=Spotlighting, Para.=Paraphrasing, Sand.=Sandwiching, S+S=Spotlight+Sandwich, P+S=Para+Spotlight.

Model	Domain	Base.	Spot.	Para.	Sand.	S+S	P+S
Haiku	Financial	26.7	13.3	6.7	13.3	6.7	6.7
	Legal	13.3	6.7	3.3	10.0	6.7	3.3
	General	3.3	0.0	3.3	3.3	0.0	0.0
Llama	Financial	33.3	43.3	13.3	33.3	36.7	16.7
	Legal	6.7	10.0	3.3	10.0	3.3	6.7
	General	26.7	16.7	13.3	23.3	20.0	6.7
Gemini	Financial	30.0	30.0	3.3	30.0	23.3	10.0
	Legal	23.3	20.0	6.7	13.3	6.7	3.3
	General	10.0	10.0	0.0	16.7	10.0	3.3

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Table 2: Utility loss breakdown for task_success=False trials. Over-refusal: agent explicitly declines. Confusion: response <50 words. Genuine failure: agent attempted but gave incorrect output. Paraphrasing uniquely achieves 0% over-refusal across all models.

Model	Defense	n_{fail}	Over-refusal	Confusion	Genuine fail
Haiku	Baseline	10	0.0%	0.0%	100.0%
	Spotlighting	7	0.0%	0.0%	100.0%
	Paraphrasing	5	0.0%	0.0%	100.0%
	Sandwiching	8	0.0%	0.0%	100.0%
	Spot.+Sandwich	10	0.0%	0.0%	100.0%
	Para.+Spotlight	16	0.0%	0.0%	100.0%
	Llama Guard 4	20	90.0%	0.0%	10.0%
Llama	Baseline	45	86.7%	0.0%	13.3%
	Spotlighting	57	75.4%	0.0%	24.6%
	Paraphrasing	23	0.0%	4.3%	95.7%
	Sandwiching	44	75.0%	2.3%	22.7%
	Spot.+Sandwich	42	83.3%	0.0%	16.7%
	Para.+Spotlight	26	7.7%	0.0%	92.3%
	Llama Guard 4	32	90.6%	0.0%	9.4%
Gemini	Baseline	50	4.0%	58.0%	38.0%
	Spotlighting	48	8.3%	52.1%	39.6%
	Paraphrasing	39	0.0%	28.2%	71.8%
	Sandwiching	48	0.0%	64.6%	35.4%
	Spot.+Sandwich	38	0.0%	55.3%	44.7%
	Para.+Spotlight	47	0.0%	31.9%	68.1%
	Llama Guard 4	32	62.5%	25.0%	12.5%

Table 3: Baseline camouflage ASR by domain (n=30 per cell). Financial is the highest-risk domain across all three models.

Model	Domain	ASR	n
Haiku	Financial	26.7%	30
	Legal	13.3%	30
	General	3.3%	30
Llama	Financial	33.3%	30
	Legal	6.7%	30
	General	26.7%	30
Gemini	Financial	30.0%	30
	Legal	23.3%	30
	General	10.0%	30

Table 4: Overall defense evaluation results across all 21 condition–model combinations. Static ASR and camouflage ASR are attack success rates; $\mathrm{CDG}_{\mathrm{def}}$ = Static ASR − Cam. ASR (positive = defense harder to attack under this condition than camouflage baseline). Utility = fraction of trials with legitimate task completed. **Bold**: lowest camouflage ASR per model. Llama Guard 4 evaluated at n=45+45; all others at n=90+90.

Model	Defense	Static ASR	Cam. ASR	$\mathrm{CDG}_{\mathrm{def}}$	Utility	n
Haiku	Baseline	0.0%	14.4%	−14.4	94.4%	90+90
	Spotlighting	0.0%	6.7%	−6.7	96.1%	90+90
	Paraphrasing	0.0%	4.4%	−4.4	97.2%	90+90
	Sandwiching	0.0%	8.9%	−8.9	95.6%	90+90
	Spot.+Sandwich	0.0%	4.4%	−4.4	94.4%	90+90
	Para.+Spot.	0.0%	**3.3%**	−3.3	91.1%	90+90
	Llama Guard 4	0.0%	11.1%	−11.1	77.8%	45+45
Llama	Baseline	21.1%	22.2%	−1.1	75.0%	90+90
	Spotlighting	15.6%	23.3%	−7.8	68.3%	90+90
	Paraphrasing	3.3%	10.0%	−6.7	87.2%	90+90
	Sandwiching	15.6%	22.2%	−6.7	75.6%	90+90
	Spot.+Sandwich	12.2%	20.0%	−7.8	76.7%	90+90
	Para.+Spot.	1.1%	**10.0%**	−8.9	85.6%	90+90
	Llama Guard 4	8.9%	24.4%	−15.6	64.4%	45+45
Gemini	Baseline	38.9%	21.1%	+17.8	72.2%	90+90
	Spotlighting	11.1%	20.0%	−8.9	73.3%	90+90
	Paraphrasing	0.0%	**3.3%**	−3.3	78.3%	90+90
	Sandwiching	17.8%	20.0%	−2.2	73.3%	90+90
	Spot.+Sandwich	4.4%	13.3%	−8.9	78.9%	90+90
	Para.+Spot.	0.0%	5.6%	−5.6	73.9%	90+90
	Llama Guard 4	6.7%	20.0%	−13.3	64.4%	45+45